

Leetcode

functional requirements :

- user should be able to solve any problem based in his preference & submit solution based on his preference,
- user should be able to participate in any coding contest based in his preference,
- user should be able to search for any problem among the existing problem based in his preference

Non-functional requirements:

- data-partitioning,
- load balancing,
- fault tolerance,
- vertical scalability,
- horizontal scalability,
- low latency ,
- high throughput ,
- CAP theorem (Here we need to prioritise Strong Consistency for rank predictions during coding contests & For search scenario we need to prioritise 99.99% uptime i.e Availability>>Consistency)

Core Entities:

- user,
- problem,
- solution,
- contest,

API :

POST:/solution

```
{
  userID,
  filemetadata
}
```

GET:/search/{problemID}

HLD :

